

Usability evaluation — participant tasks

I read each scenario and instruction verbatim, started a stopwatch when the participant began, did not coach. Encouraged think-aloud.

T1 — Filter the cohort and read a trajectory

Probes: sidebar filtering; Tab B trajectory plot.

Scenario. You want to know whether a particular gut-microbiome metabolite trends downward in female patients who later develop mild cognitive impairment, ignoring the male participants who introduce a sex confound.

Instruction.

“Using only the sidebar, filter the cohort so it shows female participants who progress from NL to MCI. Switch to Tab B — Trajectory. The active metabolite at the top of the dropdown is GLCA. Tell me whether GLCA’s mean concentration goes up or down from Timepoint -3 to Timepoint 0.”

Answer expected. Down, the slope is negative across -3 to 0.

What we measure.

- Time-to-first-fixation on the sidebar Cohort section.
- Whether the trajectory-groups checklist is located without prompting.
- Whether the slope direction is correctly read (validates chart clarity at one active cohort, full CI whiskers visible).

T2 — Discover a high-effect metabolite

Probes: cross-view linking; Tab C forest → Tab B trajectory.

Scenario. Before forming a hypothesis you want to know which of the 29 metabolites carries the strongest signal at the conversion visit (slot 0) when comparing MCI → AD converters against stable cognitively normal controls.

Instruction.

“Move to Tab C — Forest plot. Make sure the Converter group is set to MCI→AD and the Baseline is stable_NL. Find the metabolite with the largest positive effect size (Hedges’ g), click directly on its dot, switch back to Tab B — Trajectory, and tell me which metabolite the chart is now showing.”

Answer expected. Whichever metabolite tops the forest plot at slot 0 — typically GHCA or one of the secondary-bile-acid conjugates.

What we measure.

- Whether the click-to-drill affordance is discovered without prompting (the “Tip: click a metabolite dot...” caption).

- Time spent on Tab C before the click is made.
- Whether the participant verifies the metabolite name on Tab B rather than recalling it from memory.

T3 — Compare two cohorts on the same metabolite

Probes: multi-group overlay; dodging + CI suppression.

Scenario. You want to compare the GLCA trajectories of MCI → AD progressors, stable_NL controls, and stable_MCI controls side by side to see whether the progressors' rise begins before the diagnosis-change visit.

Instruction.

“Back on Tab B, untick everything in the Trajectory groups checklist, then tick MCI→AD, stable_NL, and stable_MCI. Tell me, looking only at the chart, between which two slots does the orange MCI→AD line rise most steeply?”

Answer expected. Between slot -2 and slot -1 (or slot -1 and slot 0, depending on the active metabolite).

What we measure.

- Whether the participant notices the loss of CI whiskers and comments on it.
- Whether the per-slot stats table below the chart is consulted to verify the visual reading.
- Reading time per cohort (eye-tracking: dwell on the chart vs. the table).

T4 — Check an individual patient

Probes: click-to-drill from PCA; Tab D → Tab E.

Scenario. A clinician notices an outlier patient sitting far from the central cluster in PCA space and wants to know that individual's metabolite profile across visits.

Instruction.

“Switch to Tab D — PCA state. Find the data point that sits farthest from the central cluster (any of the four colour encodings is fine, use whichever feels easiest). Click that point. The dashboard should jump you to Tab E: Individual patient view. Read me the patient's RID from the sidebar and describe whether at least one of their nine metabolites shows a clear rising trend.”

Answer expected. An RID (4-digit integer) and an honest description of one metabolite's trend.

What we measure.

- Whether the participant experiences change blindness after the automatic tab switch (eye-tracking: do their eyes track to the patient panel or remain at the PCA position?).
- Discovery of the per-patient RID dropdown.

T5 — Inspect cohort-level context

Probes: Tab A readability; KPI cards; multi-chart parsing.

Scenario. A reviewer asks how representative the study cohort is in terms of sex balance and follow-up regularity before they will accept any effect-size claims you make.

Instruction.

“Go to Tab A — Cohort information. Without reading any chart in detail, tell me: (a) the total number of participants, (b) the male-to-female ratio, (c) roughly how many months separate two consecutive visits on average, and (d) at what calendar year the sex composition of the cohort begins to shift toward more females.”

Answer expected. (a) 1,274 · (b) 719 M / 555 F · (c) $\approx$ 12 months · (d) Around 2010–2011 (the visual inflection in the stacked area).

What we measure.

- Whether the participant locates the KPI cards quickly versus the charts (eye-tracking: AOI hit-rate on the top half of Tab A).
- Whether the inter-visit-gap chart is correctly interpreted as months, not weeks.

After all five tasks

- Click Stop and Download CSV in the eye-tracking control card.
- Rename the CSV with the participant ID (e.g. usability_gaze_log_P02.csv) and move it into a per-participant folder.
- Administer the System Usability Scale questionnaire (10 items, 5-point Likert; see debrief_questions.md).
- Open the three debrief questions in debrief_questions.md and record the answers verbatim.

Pilot pass

Before running with external participants, complete the five tasks yourself once end-to-end to confirm:

- the CSV writes correctly under dashboard/usability_gaze_log.csv;
- the calibration overlay closes cleanly after the last dot;
- the click-to-drill from Tab D to Tab E fires (it depends on the PCA point's customdata);
- analyze_gaze.py runs without errors on the resulting CSV.